

Machine Learning for Biology and Health

CSCI 1851
Spring 2026

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Thursday



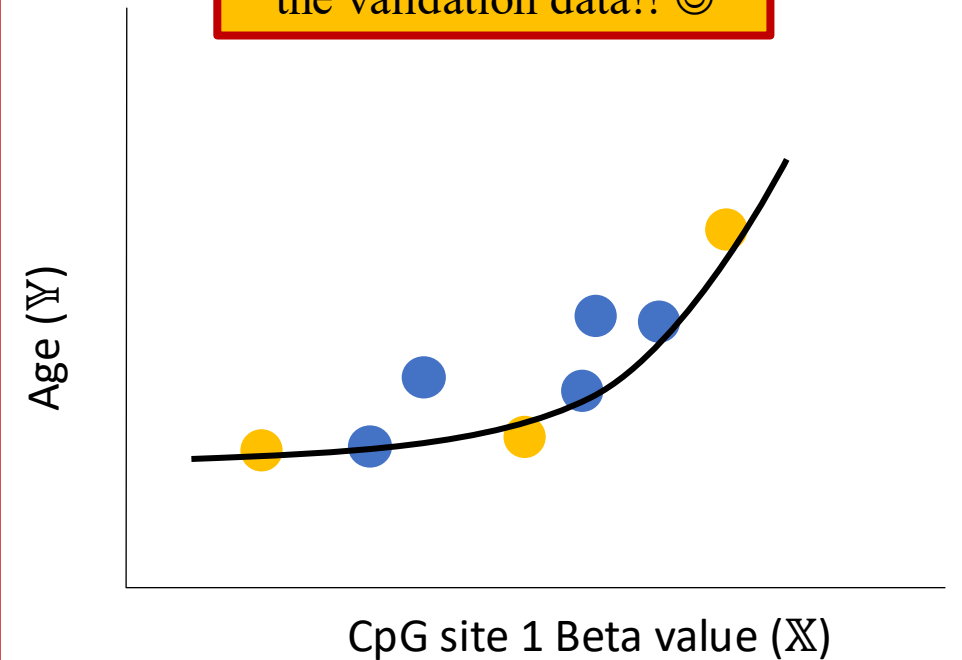
How to train your model

	Input: $\mathbb{X}$ "CpG site 1"	Target: $\mathbb{Y}$ "Age"
Training set	$x^{(1)} = 0.2$ $x^{(2)} = 0.3$ $x^{(3)} = 0.4$.	$y^{(1)} = 25$ $y^{(2)} = 35$ $y^{(3)} = 40$.
Validation set	.	.
	.	.
	.	.
	.	.
Test set	.	.
	.	.
	$x^N = \dots$	$y^N = \dots$

Cross-validation (1-fold)

Training phase

"My model is doing well on the training data AND the validation data!! 😊"



Regularization - L1 vs L2

Note:

Regularization assumes comparable scales, so standardization is important

L1 Regularization

$$\text{Cost} = \sum_{i=0}^N (y_i - \sum_{j=0}^M x_{ij} W_j)^2 + \lambda \sum_{j=0}^M |W_j|$$

Lasso Regression

L2 Regularization

$$\text{Cost} = \sum_{i=0}^N (y_i - \sum_{j=0}^M x_{ij} W_j)^2 + \lambda \sum_{j=0}^M W_j^2$$

Ridge Regression

Loss function

Regularization
Term

Handling Missing Data

- **Missing completely at random:** if the probability of missingness is the same for all units

Do nothing, let algorithm handle N/A values

Discard missing data

- **Missing at random:** probability a variable is missing depends only on available information

Imputation: Mean/Median, Random value, Most frequent value, k-nearest neighbor, Regression

- **Missingness that depends on unobserved predictors:** missingness depends on information that has not been recorded and this information also predicts the missing values

Hard case!

Today's goal - Learn about logistic regression and wrapping up discussion on linear models

(1) Logistic Regression

(2) Loss function

(3) Evaluation metrics

(4) Class Activity: Classification and Evaluation

Logistic Regression

Age prediction from DNA methylation



Input: $\mathbb{X}$

“CpG sites”

Sample	CpG ₁	CpG ₂	CpG ₃
1	0.20	0.35	0.60
2	0.30	0.45	0.65
3	0.40	0.55	0.70
4	0.50	0.65	0.75

How do I convert this to
a classification task?

Target: $\mathbb{Y}$
“Age”

➡ Function: f ➡

$f(\mathbb{X}) \rightarrow \mathbb{Y}$

Age

25

35

45

55

$\mathbb{X} \in \mathbb{R}$

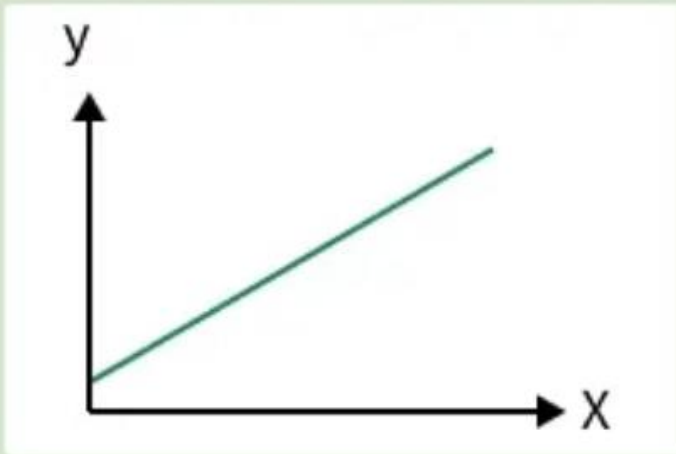
$\mathbb{Y} \in \mathbb{Z}$

(Numerical output)

Logistic regression

Linear Regression

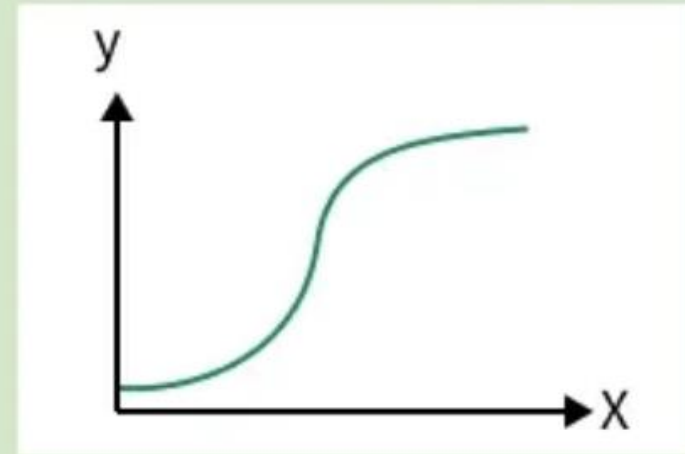
- Predicts continuous values
- Uses best-fit line
- Solves regression problems



vs

Logistic Regression

- Predicts categorical classes
- Uses sigmoid S-curve
- Solves classification problems

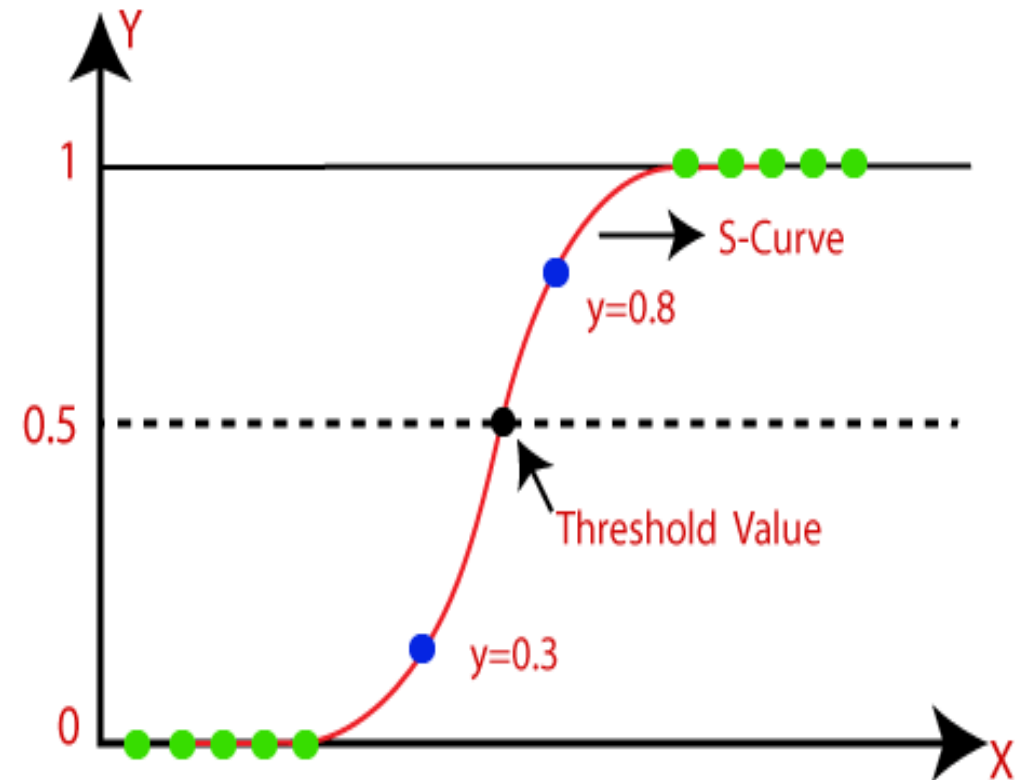


Main ingredient: Sigmoid function

1. The sigmoid converts the raw output of the model into a probability value between 0 and 1.
2. This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the sigmoid curve or logistic curve.

Because probabilities must lie between 0 and 1, the sigmoid function is perfect for this purpose.

3. In logistic regression, we use a threshold value usually 0.5 to decide the class label.
 - If the sigmoid output is same or above the threshold, the input is classified as Class 1.
 - If it is below the threshold, the input is classified as Class 0.



How does Logistic regression work?

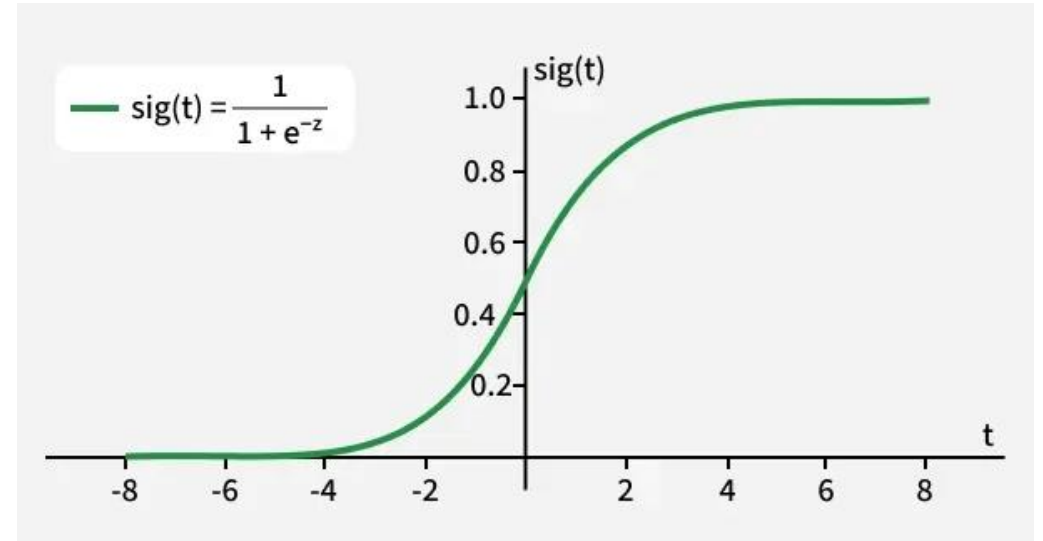
$$X = \begin{bmatrix} x_{11} & \dots & x_{1m} \\ x_{21} & \dots & x_{2m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nm} \end{bmatrix} \quad Y = \begin{cases} 0 & \text{if } \textit{Class 1} \\ 1 & \text{if } \textit{Class 2} \end{cases}$$

$$z = w \cdot X + b$$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$P(y = 1) = \sigma(z)$$

$$P(y = 0) = 1 - \sigma(z)$$



- $\sigma(z)$ tends towards 1 as $z \rightarrow \infty$
- $\sigma(z)$ tends towards 0 as $z \rightarrow -\infty$
- $\sigma(z)$ is always bounded between 0 and 1

Does this
remind you of
any distribution?

Bernoulli Distribution

A Bernoulli random variable Y has one parameter:

$$p = P(Y = 1)$$

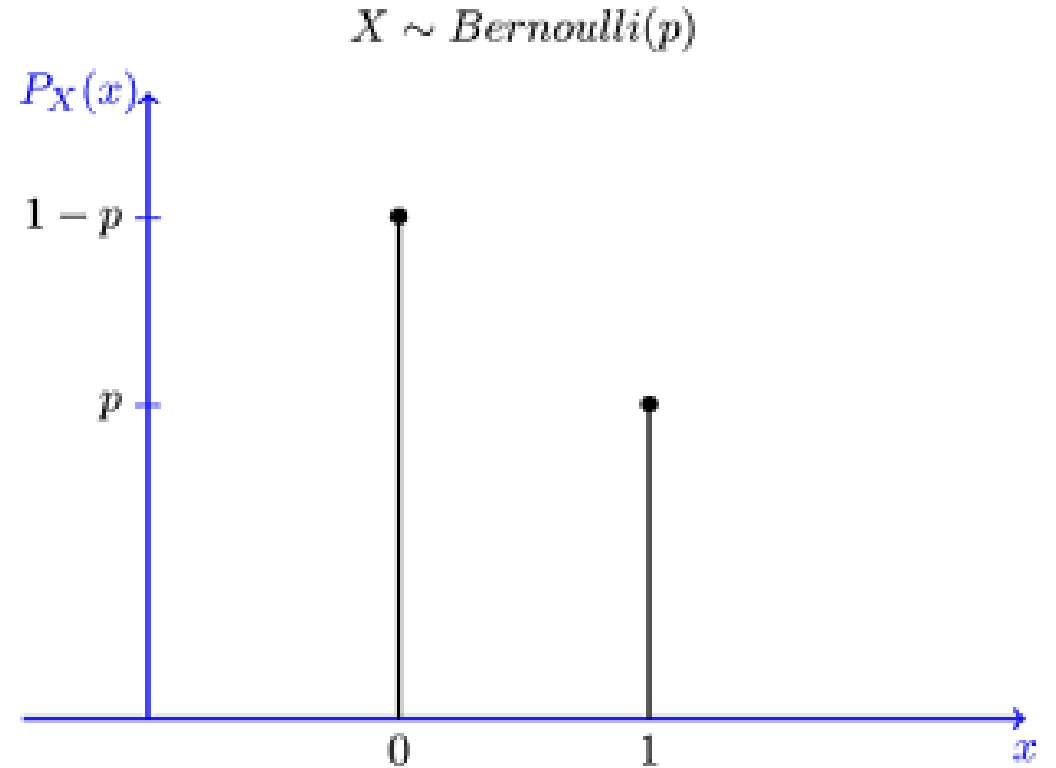
So:

- $P(Y = 1) = p$
- $P(Y = 0) = 1 - p$

$$P(Y = y) = p^y(1 - p)^{1-y}, \quad y \in \{0, 1\}$$

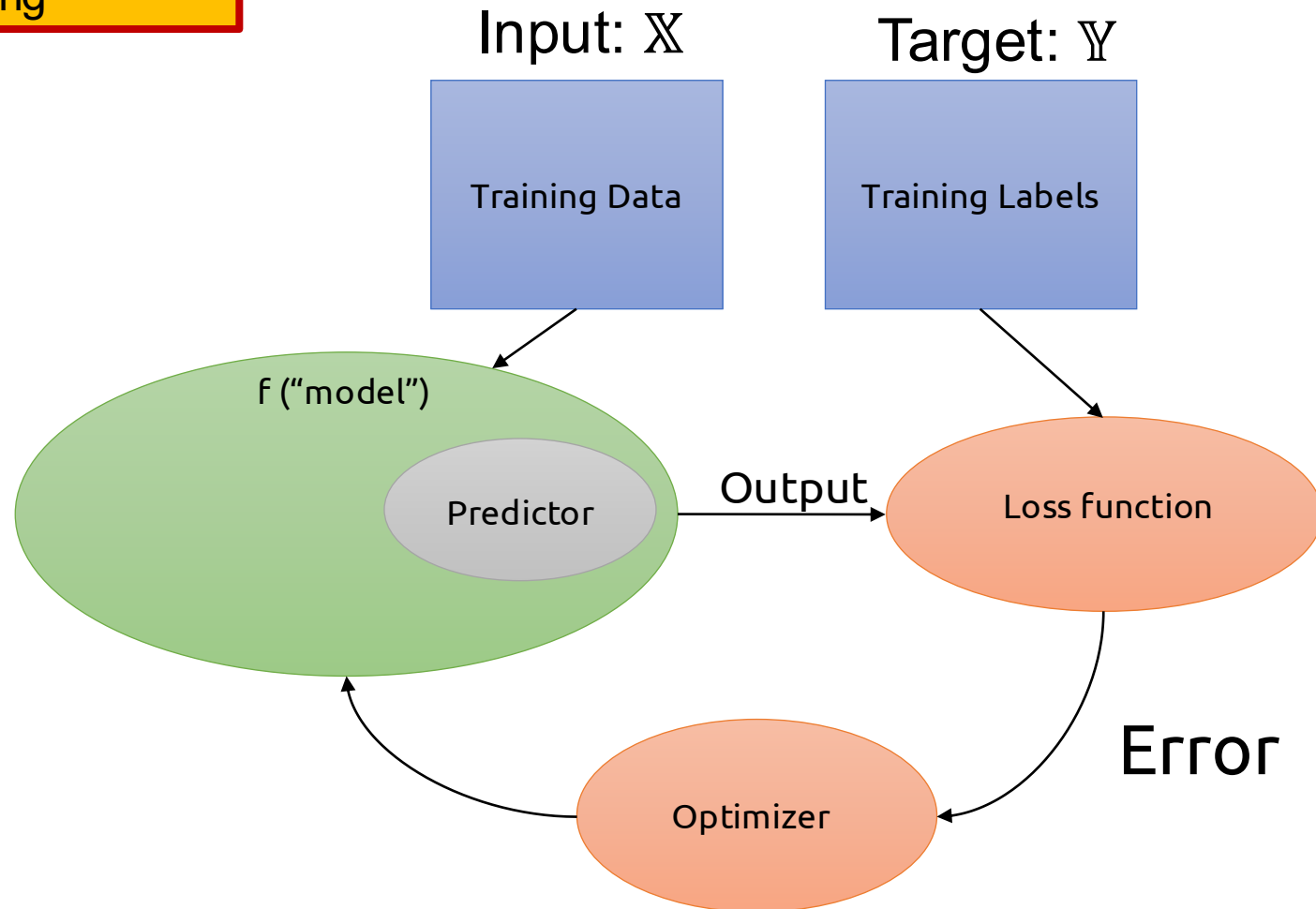
In binary classification:

- Labels $y \in \{0, 1\}$
- Each label is a Bernoulli draw
- Models predict $p = P(y = 1 \mid x)$



“Classic” Supervised Learning in Machine Learning

Training



Any questions?



Loss Function

Likelihood Function for Logistic Regression

$$L(b, w) = \prod_{i=1}^n p(x_i)^{y_i} (1 - p(x_i))^{1-y_i}$$

Cross Entropy Loss (for Binary classification)

y = true label of class (0 or 1)

p = predicted probability of class 1

$$\mathcal{L}(w, b) = \frac{1}{n} \sum_{i=1}^n [-y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i)]$$

$$-(y \log(p) + (1 - y) \log(1 - p))$$

$$y = 1, p = 0.9$$

$$y = 0, p = 0.9$$

$$y = 1, p = 0.1$$

Any questions?



Some examples:

$$\log(0.9) = -0.10$$

$$\log(0.1) = -2.3$$

We get this probability by using a Sigmoid function

$$y = 0, p = 0.1$$

Evaluation metrics

Evaluation: Regression

The diagram illustrates the Mean Absolute Error (MAE) formula with several annotations:

- A blue box around $\frac{1}{n}$ is labeled "Divide by the total number of data points".
- A green box around y is labeled "Actual output value".
- An orange box around $\hat{y}$ is labeled "Predicted output value".
- A bracket under the absolute value term $|y - \hat{y}|$ is labeled "The absolute value of the residual".
- The summation symbol Σ is labeled "Sum of".

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

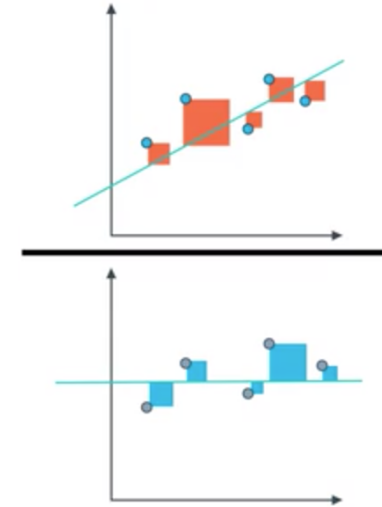
$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Evaluation: Regression

○ R2 SCORE

- BAD MODEL
The errors should be similar.
R2 score should be close to 0.
- GOOD MODEL
The mean squared error for the linear regression model should be a lot smaller than the mean squared error for the simple model.
R2 score should be close to 1.

$$R^2 = 1 -$$



Any questions?



$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Classification: Why accuracy might not be sufficient

How to classify samples?

1. Predict class probabilities
2. Calibrate a threshold

False Positive. Predict an event when there was no event.

False Negative. Predict no event when in fact there was an event.

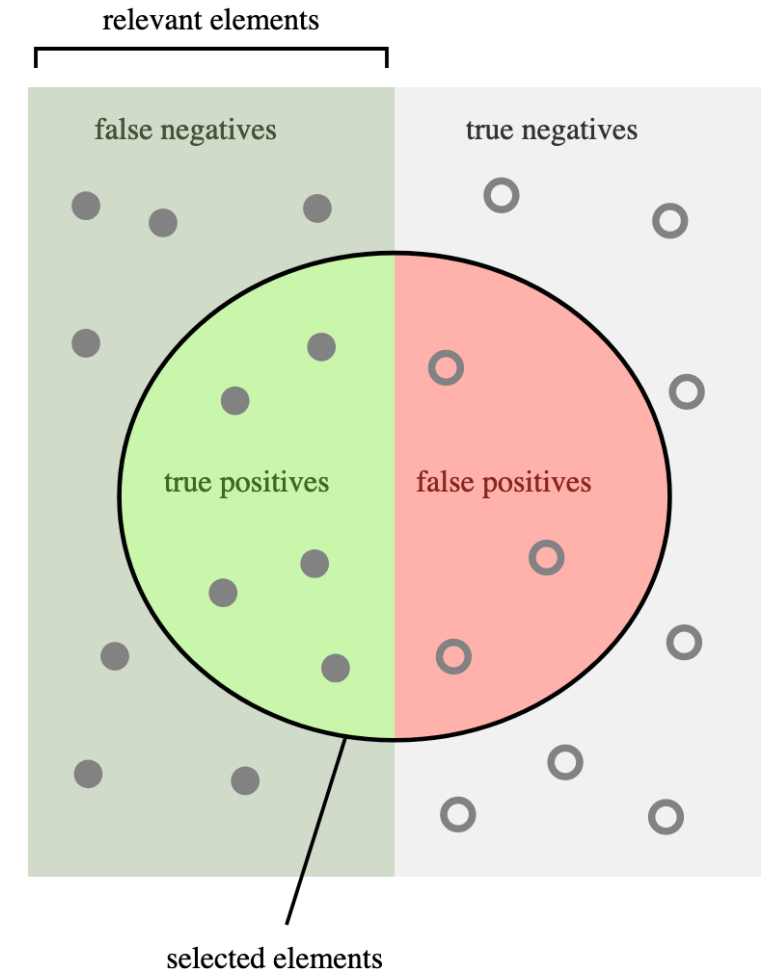
Example: For predicting patient sample as indicative of heart disease.

False positive: We predict “heart disease” but patient is fine

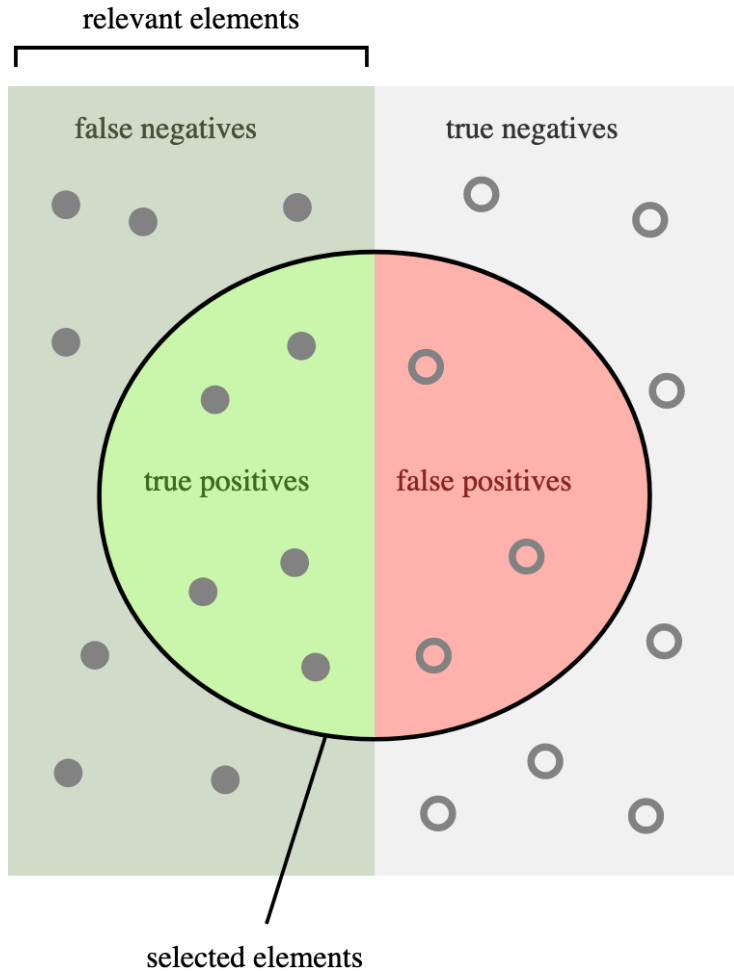
False negative: Patient has heart disease and we predict “no heart disease”

A false negative can have more severe consequence compared to a false positive

Image Courtesy: Walber - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=36926283>



Evaluation: Classification



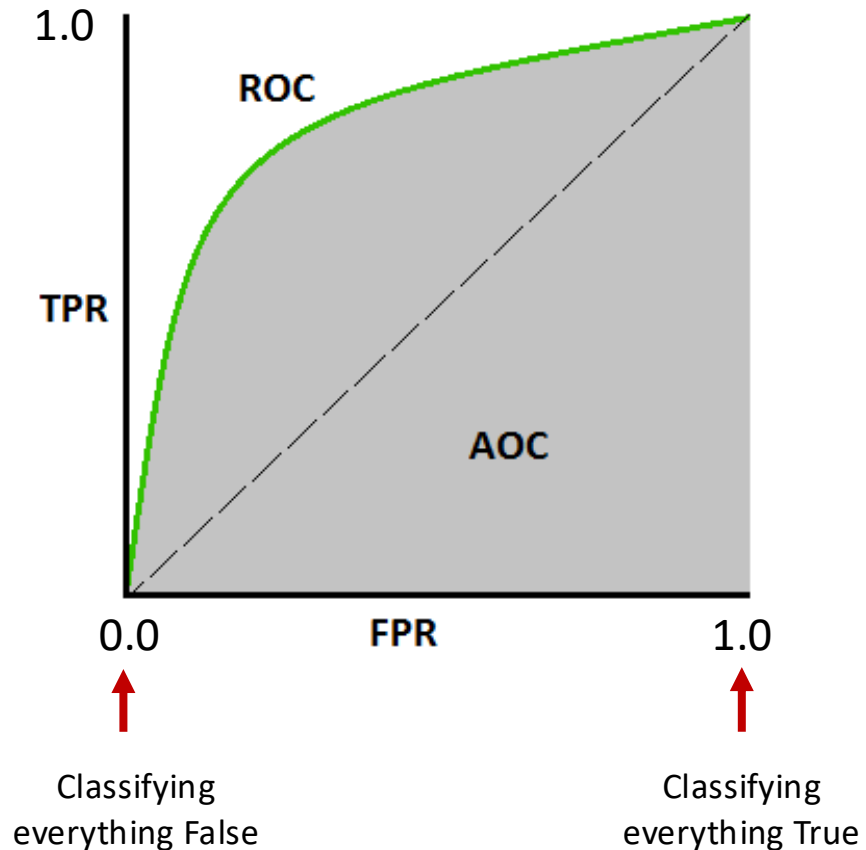
True Positive Rate = True Positives / (True Positives + False Negatives)

How good the model is at predicting the positive class when the actual outcome is positive.

False Positive Rate = False Positives / (False Positives + True Negatives)

How often a positive class is predicted when the actual outcome is negative

Evaluation: Classification (ROC Curve)

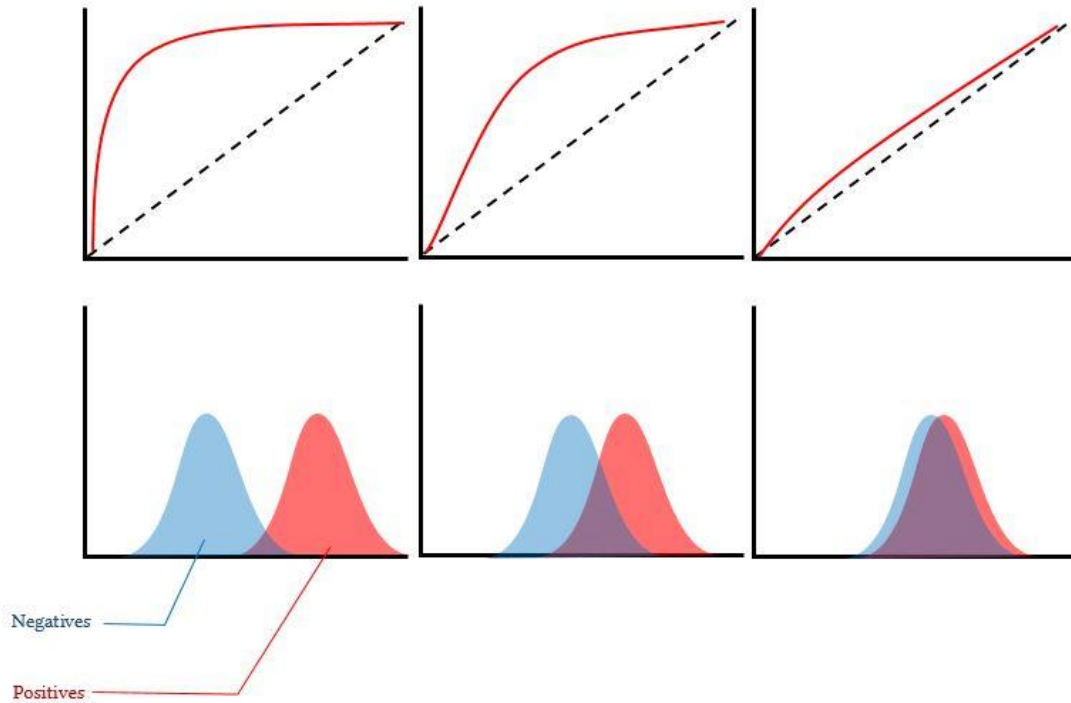


y	p
1	0.9
1	0.8
1	0.5
0	0.3
0	0.1
1	0.1

$$TPR = \frac{TP}{TP+FN}$$

$$FPR = \frac{FP}{FP+TN}$$

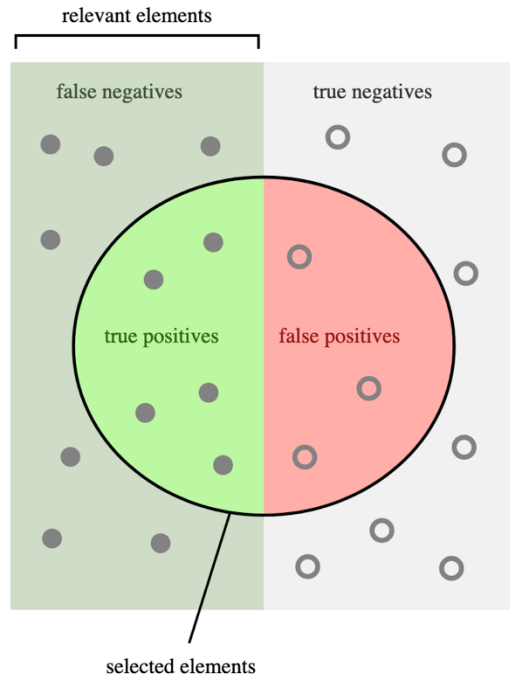
Evaluation: Classification (AUROC)



AUROC Score > 0.70 : Model will assign a higher probability to a randomly chosen real positive occurrence than a negative occurrence on average

AUROC Score ~ 0.5 : A model is randomly classifying samples at each threshold

Evaluation: Classification (Precision- Recall)



$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$
$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

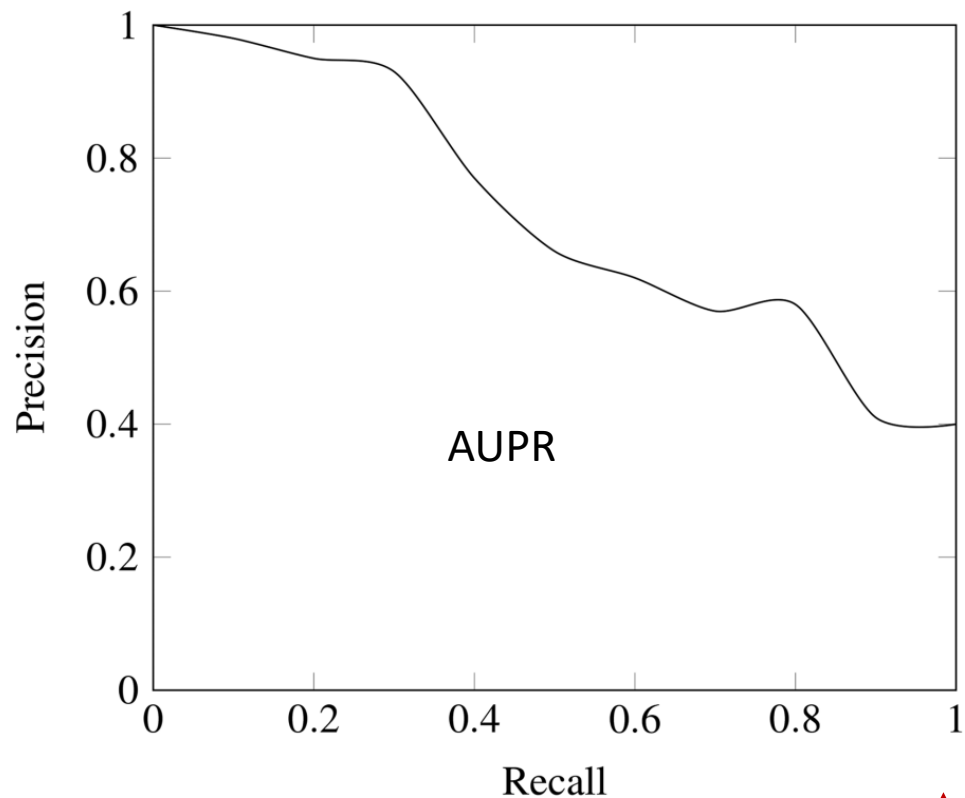
Precision = True Positives / (True Positives + False Positives)

How good a model is at predicting the positive class

Recall = True Positives / (True Positives + False Negatives)

Same as true positive rate, how good the model is at predicting the positive class when the actual outcome is positive.

Evaluation: Classification (AUPRC)



↑
0 False positives
More False negatives

↑
0 False Negatives
More False Positives

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

When dealing with rare samples:

If there are a large number of class 0 examples, we are less interested model's performance at predicting class 0 correctly, e.g. high true negatives.

Evaluation: F1-Score

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

Which metric would you prefer? F1 score/AUPRC score?

Any questions?



Class activity

Classification and Evaluation

Recap - Learn about logistic regression and wrapping up discussion on linear models

(1) Logistic Regression

(2) Loss function

(3) Evaluation metrics

(4) Class Activity: Classification and Evaluation

Homework Reading:

[Sections 7.1, 7.2, 7.3, 7.4 in CIML Book](#)

Wrap up



What was the clearest point today?

What was the muddiest point today?

